

10 Group 2

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Down Group 2, ionisation energy decreases due to increased distance and shielding. Metal reactivity with water increases. Hydroxide solubility rises down the group; sulfate solubility falls. Thermal decomposition of carbonates/nitrates becomes harder as larger cations polarise anions less strongly.

Worked example: follow the reasoning

BaSO_4 is insoluble and can identify sulfate ions as a white precipitate when Ba^{2+} is added under suitable test conditions.

Quick check · multiple choice

A How does Group 2 hydroxide solubility generally change down the group? [1]

- A Decreases
- B Increases
- C Stays zero
- D Cannot be measured

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why MgCO_3 decomposes more readily than BaCO_3 on heating. [3]

Answers and explanation

Quick check: B. Increases

Question 1 [3 marks]

Mg²⁺ is smaller [1] and more strongly polarises carbonate ion [1], weakening C–O bonding and lowering thermal stability [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.